

System Information Commands

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- <1> `uname` - Prints system information.
- <2> `uptime` - Shows how long the system has been running.
- <3> `hostname` - Displays or sets the system's hostname.
- <4> `dmesg` - Prints kernel and boot messages.
- <5> `free` - Displays memory usage.
- <6> `top` - Shows real-time system processes and resource usage.
- <7> `htop` - Interactive process viewer.
- <8> `lsb_release` - Prints distribution-specific information.
- <9> `arch` - Displays machine architecture.
- <10> `uname` - Shows system information (redundant).



File and Directory Management

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- <1> `ls` - Lists directory contents.
- <2> `cd` - Changes the current directory.
- <3> `pwd` - Prints the current working directory.
- <4> `mkdir` - Creates a new directory.
- <5> `rmdir` - Removes empty directories.
- <6> `touch` - Creates an empty file or updates the timestamp of an existing file.
- <7> `rm` - Removes files or directories.
- <8> `cp` - Copies files or directories.
- <9> `mv` - Moves or renames files or directories.
- <10> `ln` - Creates hard or symbolic links.



File Viewing and Editing

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- <1> cat - Concatenates and displays file content.
- <2> more - Views file content one screen at a time.
- <3> less - Similar to more but with more features.
- <4> tail - Outputs the last part of files.
- <5> head - Outputs the first part of files.
- <6> nano - Simple text editor in the terminal.
- <7> vi/vim - Powerful text editor.
- <8> gedit - GUI text editor for GNOME.
- <9> awk - Pattern scanning and processing language.
- <10> sed - Stream editor for filtering and transforming text.



File Permissions and Ownership

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- <1> `chmod` - Changes file permissions.
- <2> `chown` - Changes file owner and group.
- <3> `chgrp` - Changes group ownership.
- <4> `umask` - Sets default file permissions.
- <5> `stat` - Displays detailed file or file system status.
- <6> `getfacl` - Gets file access control lists.
- <7> `setfacl` - Sets file access control lists.
- <8> `lsattr` - Lists file attributes on a Linux second extended file system.
- <9> `chattr` - Changes file attributes on a Linux file system.
- <10> `usermod` - Modifies user account properties.



Networking Commands

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- <1> `ifconfig` - Configures network interfaces.
- <2> `ip` - Shows/manages IP addresses and routing.
- <3> `ping` - Checks network connectivity.
- <4> `netstat` - Displays network connections, routing tables, interface statistics, masquerade connections, and multicast memberships.
- <5> `ss` - Another utility to investigate sockets.
- <6> `traceroute` - Tracks the route packets take to a network host.
- <7> `wget` - Downloads files from the web.
- <8> `curl` - Transfers data from or to a server. second extended file system.
- <9> `nslookup` - Queries DNS to obtain domain name or IP address mapping.
- <10> `dig` - Another utility for querying DNS name servers.



Process Management

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- <1> ps - Displays current processes.
- <2> pgrep - Searches for processes by name.
- <3> pkill - Terminates processes by name.
- <4> kill - Sends a signal to a process to terminate it.
- <5> killall - Kills all processes by name.
- <6> bg - Resumes suspended jobs in the background.
- <7> fg - Brings background jobs to the foreground.
- <8> jobs - Lists active jobs.
- <9> nice - Sets process priority.
- <10> renice - Alters the priority of running processes.



Disk Management

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- <1> df - Reports file system disk space usage.
- <2> du - Estimates file space usage.
- <3> fdisk - Partition table manipulator.
- <4> mkfs - Builds a Linux file system.
- <5> mount - Mounts a file system.
- <6> umount - Unmounts a file system.
- <7> blkid - Locates/prints block device attributes.
- <8> lsblk - Lists information about block devices.
- <9> parted - Manages partition tables.
- <10> tune2fs - Adjusts tunable file system parameters on ext2/ext3/ext4 file systems.



System Monitoring and Performance

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- <1> top - Displays Linux tasks.
- <2> htop - Interactive process viewer.
- <3> iostat - Reports CPU and I/O statistics.
- <4> vmstat - Reports virtual memory statistics.
- <5> sar - Collects, reports, or saves system activity information.
- <6> free - Displays memory usage.
- <7> mpstat - Reports CPU usage.
- <8> uptime - Shows system uptime and load.
- <9> dstat - Versatile resource statistics.
- <10> glances - Cross-platform system monitoring tool.



User Management

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- <1> useradd - Creates a new user or updates default new user information.
- <2> userdel - Deletes a user account and related files.
- <3> usermod - Modifies a user account.
- <4> passwd - Changes a user's password.
- <5> chage - Changes user password expiry information.
- <6> groups - Shows group memberships.
- <7> groupadd - Creates a new group.
- <8> groupdel - Deletes a group.
- <9> newgrp - Logs in to a new group.
- <10> su - Switches to another user account.



Compression and Archiving

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- <1> `bzip2` - Compresses files using the Burrows-Wheeler block sorting text compression algorithm.
- <2> `bunzip2` - Command decompresses files compressed by `bzip2`.
- <3> `xz` - Command compresses files using the LZMA/LZMA2 algorithms.
- <4> `unxz` - Command decompresses files compressed by `xz`.
- <5> `tar` - Command creates or extracts tar archives.
- <6> `zip` - It packages and compresses files.
- <7> `unzip` - For extracting files from a zip archive.
- <8> `lzma` - Command compresses files using the Lempel-Ziv-Markov chain algorithm.
- <9> `lzop` - Command compresses files using the LZO algorithm, which is faster but less efficient than `gzip`.
- <10> `unlzop` - It decompresses files compressed by `lzop`.



Final Note



Understanding and mastering these 100 Linux commands will significantly improve your ability to navigate, manage, and secure a Linux system efficiently.

Further Reading and Resources

Explore our Linux-based [FAQs](#), and [blogs](#) to get a detailed insights about this operating system's operation.

Final Tips for Linux Users

- <1> Practice regularly to become proficient.
- <2> Explore and experiment with command options to understand their full capabilities.
- <3> Stay updated with the latest Linux developments and community best practices.

By familiarizing yourself with these commands, you can enhance your productivity and effectiveness as a Linux user or administrator.

Happy learning!

